

PROJECT SYNOPSIS

AskData AI – Intelligent Data Analyst Assistant using NLP, RAG and LLM

Degree: B.Sc. Data Science | Project Type: Final Year Project

1. Introduction

Data analysis often requires knowledge of SQL, spreadsheets, programming and visualization tools. AskData AI proposes an intelligent Data Analyst Assistant that allows users to upload CSV or Excel datasets and ask questions in natural language. It combines traditional data analytics with Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), data visualization and business intelligence techniques.

2. Problem Statement

Non-technical users may find it difficult to extract insights from large datasets because existing dashboards are often predefined and traditional analysis requires SQL or programming knowledge. The proposed system aims to provide a simple conversational interface that can understand analytical questions, retrieve relevant information, perform analysis and present understandable results.

3. Aim

To develop an AI-powered Data Analyst Assistant that enables users to interact with structured datasets using natural-language questions and receive data-backed answers, visualizations, KPIs and business insights.

4. Objectives

- Allow users to upload CSV and Excel datasets.
- Automatically inspect and profile uploaded data.
- Identify common data-quality issues such as missing values, duplicates and data types.
- Use NLP/query-understanding techniques for natural-language analytical questions.
- Use embeddings and vector search to retrieve relevant context.
- Use RAG to ground LLM responses in available data/context.
- Generate understandable explanations and business insights.
- Generate suitable charts and KPIs.
- Provide an interactive Streamlit interface.
- Provide report/export functionality as an extension.

5. Proposed System

After a dataset is uploaded, the system profiles and prepares the data for analysis. When a user asks a question, the application interprets the query, retrieves relevant information, performs the required analytical operation and uses an LLM to formulate a grounded response. Where appropriate, the system also displays a visualization and business interpretation.

6. System Workflow

Step	Process
1	Upload CSV / Excel
2	Data cleaning and profiling

3	Create searchable representations / embeddings
4	Store information for retrieval
5	User asks a natural-language question
6	NLP / query understanding
7	Retrieve relevant data or context
8	Perform analytical operation
9	LLM generates grounded explanation
10	Display answer, KPI and visualization

7. Technologies / Tools

Area	Technology
Programming	Python
Data Analysis	Pandas, NumPy
Database / Querying	SQL, SQLite / MySQL
NLP	NLTK / spaCy / Sentence Transformers
LLM	OpenAI-compatible API or suitable open-source LLM
RAG	LangChain or equivalent framework
Vector Search	FAISS / ChromaDB
Visualization	Plotly / Matplotlib
Application	Streamlit
Version Control	Git and GitHub

8. Major Use Cases

- Sales and revenue analytics – trends, top products and regional performance.
- Customer analytics – high-value customers and purchasing patterns.
- Marketing analytics – campaign performance and conversion analysis.
- Financial analytics – revenue, profit, expenses and KPIs.
- HR analytics – employee trends and attrition analysis.
- E-commerce analytics – orders, products and category performance.
- Customer review analytics – sentiment, complaints and common topics.
- Inventory analytics – stock levels and fast-moving products.
- Logistics and transportation analytics – delays, cancellations and locations.
- Education analytics – marks, attendance and subject performance.
- Banking and insurance analytics – customer segments, loans and claims.
- Website and social-media analytics – traffic and engagement.
- Domain-independent analysis – different datasets can be uploaded to the same system.

9. Expected Outcomes

A working prototype through which users can upload datasets, ask analytical questions in natural language, obtain data-backed answers, view visualizations and receive concise business insights.

10. Innovation / Uniqueness

Unlike a conventional static dashboard, AskData AI is designed as a conversational analytics system. It combines natural-language interaction, retrieval, LLM-based explanation and automated visualization, making the system adaptable to multiple business domains.

11. Future Scope

Natural-language-to-SQL, voice-based questions, root-cause analysis, recommendations, forecasting, anomaly detection, multi-table analysis, automated PDF/PowerPoint reports and cloud deployment.

12. Conclusion

AskData AI bridges traditional Data Analytics and Generative AI. By integrating NLP, RAG and LLM technologies with Python-based analytics and visualization, the system can make data exploration more accessible to non-technical users while providing useful analytical insights.

13. Keywords

Data Analytics, NLP, Large Language Model, RAG, Generative AI, Embeddings, Vector Database, Python, SQL, Data Visualization, Streamlit, Business Intelligence.